

# Exercise: LCA of 200 Cups of Coffee

Work through this exercise step by step. The setup is given; the calculations are yours to do. A “Check your answers” section is at the very end — try not to look until you’re done.

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## Functional Unit

**200 cups of coffee served**

A coffee shop wants to know the total CO<sub>2</sub> released and the total water extracted from the ground to serve 200 cups of coffee, tracing back through the roasting process.

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## Process Tree

The system has two processes arranged as a simple chain, just like the electricity example:

| Process                         | Reference output      | Technosphere inputs              |
|---------------------------------|-----------------------|----------------------------------|
| <b>P1 — Coffee roasting</b>     | 0.1 kg roasted coffee | —                                |
| <b>P2 — Brewing and serving</b> | 1 cup of coffee       | 0.01 kg roasted coffee (from P1) |

P1 roasts green beans (which come from nature) into roasted coffee. P2 uses that roasted coffee to brew and serve one cup. All roasted coffee stays inside the system — none of it leaves without being brewed.

## Elementary flows

| Process                  | Flow  | Type                   | Rate            |
|--------------------------|-------|------------------------|-----------------|
| P1 — Coffee roasting     | Water | Extraction from ground | 0.20 L per run  |
| P1 — Coffee roasting     | CO    | Emission to air        | 0.05 kg per run |
| P2 — Brewing and serving | CO    | Emission to air        | 0.02 kg per run |

## Process graph (structure)

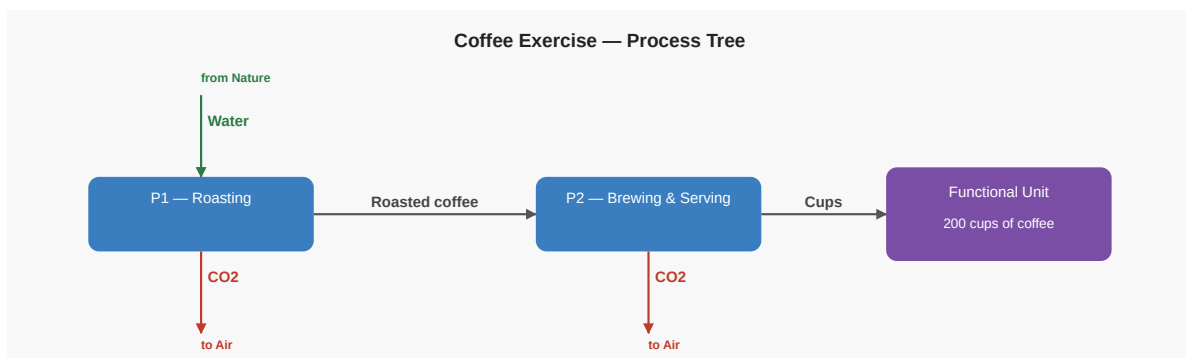


Figure 1: Structure process tree

## Characterization factors

To convert inventory flows into impact scores, use these factors:

| Flow  | Impact category | Factor                     |
|-------|-----------------|----------------------------|
| CO    | Climate change  | 1.0 kg CO -eq per kg CO    |
| Water | Water scarcity  | 0.5 points per litre water |

## Your Task

Work through each step below. Write your answers in the spaces provided.

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### Step 0a — How many runs does each process need?

We need **200 cups**. Work backwards through the chain.

**P2** makes \_\_\_\_\_ cup(s) per run. We need 200 cups, so:

$$s_2 = \frac{200 \cancel{\text{cups}}}{\text{_____} \cancel{\text{cups}}/\text{run}} = \text{_____ runs}$$

**P2 running at that level** consumes \_\_\_\_\_ kg of roasted coffee in total. P1 makes \_\_\_\_\_ kg per run, so:

$$s_1 = \frac{\text{_____} \cancel{\text{kg}}}{\text{_____} \cancel{\text{kg}}/\text{run}} = \text{_____ runs}$$

**Check — does supply meet demand?**

| Flow           | Produced              | Consumed                          | Balance |
|----------------|-----------------------|-----------------------------------|---------|
| Cups           | _____ × _____ = _____ | delivered to functional unit      | ?       |
| Roasted coffee | _____ × _____ = _____ | P2 consumes _____ × _____ = _____ | ?       |

### Process graph (scaled)

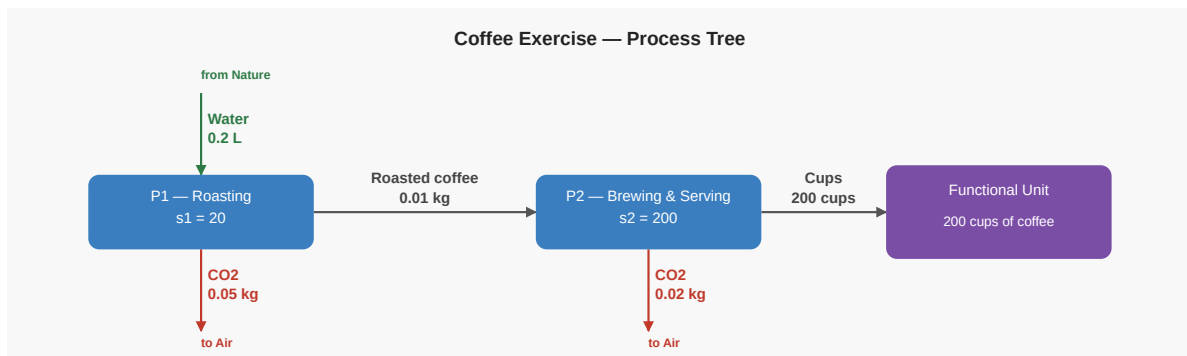


Figure 2: Scaled process tree

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**Step 0b — Calculate total emissions and extractions**

Multiply each process's rate by the total quantity it handles.

**P1 — Coffee roasting** (runs \_\_\_\_\_ times):

$$\text{Water} = 0.20 \frac{\text{L}}{\text{run}} \times \text{_____ runs} = \text{_____ L}$$

$$\text{CO}_2 = 0.05 \frac{\text{kg}}{\text{run}} \times \text{_____ runs} = \text{_____ kg}$$

**P2 — Brewing and serving** (runs \_\_\_\_\_ times):

$$\text{CO}_2 = 0.02 \frac{\text{kg}}{\text{run}} \times \text{_____ runs} = \text{_____ kg}$$

**Add across both processes:**

| Elementary flow    | From P1  | From P2  | <b>Total</b>    |
|--------------------|----------|----------|-----------------|
| Water (extraction) | _____ L  | —        | _____ <b>L</b>  |
| CO (emission)      | _____ kg | _____ kg | _____ <b>kg</b> |

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**Step 0c — Apply characterization factors**

**Climate change** (caused by CO<sub>2</sub>):

$$\text{Climate} = 1.0 \frac{\text{CO}_2\text{-eq}}{\text{kg CO}_2} \times \text{_____ kg CO}_2 = \text{_____ kg CO}_2\text{-eq}$$

**Water scarcity** (caused by water extraction):

$$\text{Water scarcity} = 0.5 \frac{\text{pts}}{\text{L}} \times \text{_____ L} = \text{_____ pts}$$

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### Step 1 — Build the technology matrix $A$

Columns = processes (P1, P2). Rows = products (roasted coffee, cups). Positive = produced, negative = consumed.

$$A = \begin{pmatrix} \_\_\_ & \_\_\_ \\ \_\_\_ & \_\_\_ \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Roasted coffee (kg)} \\ \leftarrow \text{Cups} \end{array}$$

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### Step 2 — Set up the demand vector $f$ and solve for $s$

$$f = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Roasted coffee leaving the system (kg)} \\ \leftarrow \text{Cups leaving the system} \end{array}$$

- $f = \_\_\_\_\_\_$  — we want 200 cups delivered to customers.
- $f = \_\_\_\_\_\_$  — all roasted coffee goes straight into brewing; none leaves the system.

$$f = \begin{pmatrix} \_\_\_\_\_\_ \\ \_\_\_\_\_\_ \end{pmatrix}$$

Solve  $A \cdot s = f$  by back-substitution:

$$\text{Row 2 (cups): } \_\_\_\_\_\_ \times s_2 = \_\_\_\_\_\_ \quad \Rightarrow \quad s_2 = \_\_\_\_\_\_$$

$$\text{Row 1 (roasted coffee): } \_\_\_\_\_\_ \times s_1 + \_\_\_\_\_\_ \times \_\_\_\_\_\_ = 0 \quad \Rightarrow \quad s_1 = \_\_\_\_\_\_$$

$$s = \begin{pmatrix} \_\_\_\_\_\_ \\ \_\_\_\_\_\_ \end{pmatrix}$$

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| Process | Flow | Database rate | Outputs per run | <b>Per-run rate</b> |
|---------|------|---------------|-----------------|---------------------|
|---------|------|---------------|-----------------|---------------------|

### Step 3 — Build the intervention matrix **B**

Convert each database rate (per product unit) to a per-run rate, then fill in the matrix.

| Process | Flow  | Database rate | Outputs per run | <b>Per-run rate</b> |
|---------|-------|---------------|-----------------|---------------------|
| P1      | CO    | 0.05 kg / run | —               | <b>0.05</b>         |
| P1      | Water | 0.20 L / run  | —               | <b>0.20</b>         |
| P2      | CO    | 0.02 kg / run | —               | <b>0.02</b>         |

$$B = \begin{pmatrix} \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Water (L)} \\ \leftarrow \text{CO}_2 \text{ (kg)} \end{array}$$

(columns left to right:  $P_1, P_2$ )

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### Step 4 — Compute the inventory vector $R = B \cdot s$

$$R[\text{Water}] = (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ L}$$

$$R[\text{CO}_2] = (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ kg}$$

$$R = \begin{pmatrix} \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Water (L)} \\ \leftarrow \text{CO}_2 \text{ (kg)} \end{array}$$

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**Step 5 — Apply the characterization matrix C, get impact vector h = C · R**

$$C = \begin{pmatrix} 0 & 1.0 \\ 0.5 & 0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Climate change (kg CO}_2\text{-eq)} \\ \leftarrow \text{Water scarcity (pts)} \end{array}$$

(columns left to right: Water, CO<sub>2</sub>)

$$h[\text{Climate}] = (\text{ } \times \text{ }) + (\text{ } \times \text{ }) = \text{ } \text{ kg CO}_2\text{-eq}$$

$$h[\text{Water scarcity}] = (\text{ } \times \text{ }) + (\text{ } \times \text{ }) = \text{ } \text{ pts}$$

$$h = \begin{pmatrix} \text{ } \\ \text{ } \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Climate change (kg CO}_2\text{-eq)} \\ \leftarrow \text{Water scarcity (pts)} \end{array}$$

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**Check Your Answers**

Scaling vector: - s<sub>1</sub> = 20 runs - s<sub>2</sub> = 200 runs

Inventory (R): - CO : 5.0 kg - Water: 4.0 L

Impacts (h): - Climate change: 5.0 kg CO<sub>2</sub>-eq - Water scarcity: 2.0 pts

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**Full Matrix Solution**

**Step 1 — Technology Matrix A**

$$A = \begin{pmatrix} 0.1 & -0.01 \\ 0 & 1 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Roasted coffee (kg)} \\ \leftarrow \text{Cups} \end{array}$$

(columns left to right: P<sub>1</sub>, P<sub>2</sub>)

## Step 2 — Demand Vector $f$ and Scaling Vector $s$

$$f = \begin{pmatrix} 0 \\ 200 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Roasted coffee (kg)} \\ \leftarrow \text{Cups} \end{array}$$

Solving  $A \cdot s = f$  by back-substitution:

$$s = \begin{pmatrix} 20 \\ 200 \end{pmatrix} \quad \begin{array}{l} \leftarrow P_1 \\ \leftarrow P_2 \end{array}$$

## Step 3 — Intervention Matrix $B$

$$B = \begin{pmatrix} 0.20 & 0 \\ 0.05 & 0.02 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Water (L)} \\ \leftarrow \text{CO}_2 \text{ (kg)} \end{array}$$

(columns left to right:  $P_1$ ,  $P_2$ )

## Step 4 — Inventory Vector $R = B \cdot s$

$$R = B \cdot s = \begin{pmatrix} 0.20 & 0 \\ 0.05 & 0.02 \end{pmatrix} \begin{pmatrix} 20 \\ 200 \end{pmatrix} = \begin{pmatrix} 4.0 \\ 5.0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Water (L)} \\ \leftarrow \text{CO}_2 \text{ (kg)} \end{array}$$

## Step 5 — Characterization Matrix $C$ and Impact Vector $h = C \cdot R$

$$C = \begin{pmatrix} 0 & 1.0 \\ 0.5 & 0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Climate change (kg CO}_2\text{-eq)} \\ \leftarrow \text{Water scarcity (pts)} \end{array}$$

(columns left to right:  $Water$ ,  $CO_2$ )

$$h = C \cdot R = \begin{pmatrix} 0 & 1.0 \\ 0.5 & 0 \end{pmatrix} \begin{pmatrix} 4.0 \\ 5.0 \end{pmatrix} = \begin{pmatrix} 5.0 \\ 2.0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Climate change (kg CO}_2\text{-eq)} \\ \leftarrow \text{Water scarcity (pts)} \end{array}$$

## Summary

To produce **200 cups of coffee**: - Extract **4.0 L of water** from the ground - Emit **5.0 kg of CO** to the atmosphere - Climate change impact: **5.0 kg CO -eq** - Water scarcity impact: **2.0 points**